Learning to Play the Pokémon Trading Card Game: Imitation vs Neural-Guided Search

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Abstract

This report studies agents for the Pokémon Trading Card Game (Kaggle *PTCG AI Battle Challenge*), with chance, hidden cards and a move set that changes at every decision. The first generation stacked imitation, offline RL, self-play PPO and weight ensembling on a relational transformer. It matched 63% of expert moves but never rated above 650 on the ladder. The second generation uses a compact transformer as prior and value for determinized Monte Carlo tree search on the exact simulator, improves it with self-play and expert iteration, and evaluates it with confidence intervals and sequential tests. Search lifts the gauntlet score from 0.75 to 0.90 and the ladder rating μ to ≈770. A learned belief over the opponent's hand predicts hidden cards better than counting, yet it does not improve play: one deep tree beats several shallow determinized ones.

1. Introduction

Each Pokémon TCG player pilots a 60-card deck. Hands, deck order and prize cards are hidden, draws and coin flips are random, and the legal moves change at every decision (play, attach, evolve, attack, target, pick k of n). Kaggle provides 5k high-rated replays, a C++ simulator that can step a game forward once the hidden cards are guessed, and a rated ladder. Agents get 2 CPUs and 600 s per game.

Two generations of agents are presented. The first (§2) tests how far learning alone goes. The second (§3), built after its failures, asks what a learned model gains from planning with the real engine. The contributions are an account of which offline gains carried over to play, a search agent that lifts the rating from 650 to ≈770, controlled ablations (belief, determinization, budget, expert iteration) and an evaluation protocol that, unlike the first one, agreed with the ladder where tested. Code: https://github.com/AntoCordeiro/ptcg-agent.

Related work. AlphaGo and AlphaZero pair policy/value networks with PUCT search (Silver et al., 2016; 2018); expert iteration distils the search back into the network (Anthony et al., 2017). Under hidden information, perfect-information Monte Carlo samples *determinizations* of the hidden state (Long et al., 2010), while information-set MCTS keeps one tree across samples (Cowling et al., 2012). Other ingredients are transformers with pointer outputs (Vaswani et al., 2017; Vinyals et al., 2019), IQL (Kostrikov et al., 2022), PPO (Schulman et al., 2017) and WiSE-FT (Wortsman et al., 2022).

2. First generation: learning without planning

Model. *PTCGDualCrossNet* reads 60 state tokens and 80 option tokens (332-d hand-crafted features plus frozen card-text embeddings) and a global vector holding a card-tracker belief. A 4-layer encoder (d=256) feeds 3 cross-attention layers in which options attend to the state, with biases from 8 symbolic relations such as "evolves from". Gated self-attention compares options; an autoregressive pointer handles k-of-n choices.

Training. Behavioural cloning (BC) used 75k decisions from the top ≈650 replays, followed by IQL (expectile τ=0.75, advantage-weighted extraction), league PPO against past snapshots over 5–15 decks, and WiSE-FT interpolation of the BC, IQL and PPO weights. A light search took lethal attacks, vetoed deck-outs and re-ranked close ties, at 6 ms per move.

Findings. The largest gain came from fixing a data bug (observations paired with the action one step early): BC accuracy rose from 40.9% to 63.3% (IQL: 63.4%). None of the rest reached the board: an agent rated +240 Elo *better* in a 40-game local test scored 500 on the ladder, against 650 for its predecessor, and over 350 local games the 650-rated

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Deep Learning and Applied AI 2026, Sapienza University of Rome, 2nd semester a.y. 2025/2026.

Table 1. System comparison. Expert-move accuracy is measured on each generation's own validation/test split. Gauntlet: score vs the six fixed opponents (95% CI). Ladder: Kaggle rating μ (start 600).

System	top-1	gauntlet	ladder
Gen-1 BC / IQL	63.3 / 63.4%	–	–
Gen-1 WiSE + light search	–	–	650
Gen-1 final (search v4)	–	–	500
Gen-2 policy only	70.0%	.75 [.69,.80]	–
Gen-2 MCTS, 4 worlds, 2 s	–	.90 [.85,.94]	737
Gen-2 MCTS, 1 world, 2 s	–	–	768
+ ExIt, 2–3 s, deck library	–	–	**770**

agent scores only 0.32. Small head-to-head tests between agents of the same project did not predict the ladder.

3. Second generation: planning with the simulator

Network. Entity/option tokens stay, hand-crafted features go; option tokens add the embeddings of the cards they act on. A 4-layer pre-norm transformer (d=192, 2.6M parameters) feeds four heads: policy, a count head for k-of-n choices, value, and a *belief* head that predicts the opponent's hand, supervised by the opponent's own view stored in the replay. Training uses 453k decisions from the top 3,000 replays, split 90/5/5 by game. With only 3k game outcomes the value head overfits, so it is fine-tuned on 8k self-play games while the policy stays anchored on replays.

Search. At each decision K determinizations are sampled (own deck/prizes uniform; opponent decklist the most consistent of 120 lists mined from replays and ladder games; opponent hand uniform or from the belief head). PUCT then runs on the simulator with network priors and value leaves. The policy plays forced moves; root visits are summed over worlds. A time manager spends $\min(c, (T_{\text{left}} - r)/q)$ seconds per decision. Expert iteration (ExIt) fine-tunes the net on MCTS self-play visit counts.

Protocol. Offline metrics use held-out games; online, candidates face a fixed gauntlet of six opponents (two first-generation agents and the Gen-2 policy piloting four meta decks) under a simulated time budget, reported with 95% Wilson intervals, paired head-to-head ablations and an SPRT (H_0: 0, H_1: +30 Elo).

4. Results

Offline and deck. The Gen-2 policy predicts 70.0% of expert moves (uniform: 21.7%). The belief head beats counting even when counting knows the true decklist, and

Table 2. Gen-2 offline results (held-out games) and online ablations (score of A vs B, 95% CI).

Policy top-1, main phase $\geq$5 options (uniform)	60.7% (11.6%)
Value Brier: replay-only / +self-play (const.)	.254 / .226 (.249)
Belief AUROC: learned / count w/ true list	.752 / .711
MCTS-64 vs own policy (200 games)	.59 [.52, .66]
learned vs count-based hand sampling (400)	.48 [.43, .53]
4 worlds vs 1 world, equal sims (300)	.39 [.34, .45]
ExIt-2 + MCTS vs base + MCTS (300)	.57 [.51, .62]
ExIt-3 + MCTS vs ExIt-2 + MCTS (300)	.50 [.45, .56]
searching k-of-n choices vs policy (400)	.48 [.43, .53]

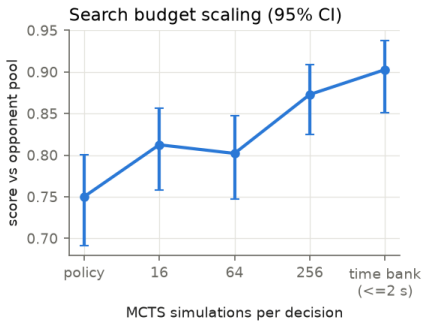


Figure 1. Gauntlet score vs search budget (Crustle deck, 180–240 games per point, 95% CI); "time bank" is the deployed agent.

self-play makes the value better than a constant (Tab. 2). Dragapult, the top deck in replays, scores only 0.49 with this policy, Kangaskhan/Crustle 0.73 (240 games each).

Search. Most of the strength comes from search (Fig. 1): 0.75 for the raw policy, 0.87 at 256 simulations and 0.90 for the deployed agent, reproduced on a second seed. Belief-sampled hands do not help, and four determinizations lose to one deeper tree (-75 Elo); at this depth the exact hand rarely matters. ExIt with 64-simulation targets leaves the raw policy unchanged but makes search stronger (+49 Elo). A third round, and searching k-of-n choices, gave nothing.

Ladder. Gen-2 agents rate 737–770, Gen-1 agents 500–650 (Tab. 1). The one local ranking checkable there ($1 > 4$ worlds) held (768 vs 737).

5. Discussion and conclusions

Learning alone plateaued: richer models and RL improved imitation, not play, while a smaller network guiding search on the true engine gained $\approx$120 rating points. Belief modelling helps prediction, not play. Limitations: four of six gauntlet opponents share the agent's network and chose its deck; the gauntlet saturates (0.90) while 800+-rated opponents remain hard (0.36–0.52).

6. Statement on the use of AI

I used AI assistants throughout the project, in two phases.

First generation (§2). My coding assistant was Gemini 3.8 Flash (Google). It wrote a large part of the code (replay parsing, feature extraction, the relational transformer, the BC/IQL/PPO training scripts, the light search and the deck tournament), and it helped me debug the pipeline and go through Kaggle replays of lost games.

Second generation (§3) and report. I then moved to Claude Opus 5.5 (Anthropic), working as an agentic assistant. It reviewed the first system and helped me plan the restart, wrote most of the new code (parser, network, MCTS, evaluation harness, submission builder), ran training and evaluation jobs on my machine through a job queue, analysed logs and ladder episodes, made the figures and helped me draft and edit this report.

The goals and the main design decisions were mine: what to keep, what to drop, which agents to submit. I ran the Kaggle submissions, reviewed the code and checked every reported number against the logged outputs in the repository (`experiments/`, `RESULTS.md`). I understand the methods and the code, can explain them, and take full responsibility for the content of this report.

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Appendix A: first-generation details

Table 3. Gen-1 local Bradley–Terry ladder (mirror deck, 20 games per pair; Elo relative to the mean). Note the small samples; the top-rated agents later scored lower on the ladder than `wise`.

Agent	BT Elo	mirror win rate
self-play PPO v3	+180.5	70.6%
search v4 (WiSE v3)	+120.7	62.5%
WiSE v3 (IQL + PPO + IQL-rel.)	+120.7	62.5%
IQL v3	+116.1	61.9%
IQL relational	+111.6	61.3%
BC	+107.1	60.6%
WiSE v1 (ladder 650)	+44.0	51.9%
heuristic	−296.1	15.6%
first submission (ladder 250–400)	−504.6	3.1%

Other Gen-1 components: a championship-deck tournament over decks reconstructed from top replays (16 games per pair, which selected Hydrapple/Ogerpon); a card tracker with archetype inference; rules for multi-select counts (e.g. attaching 2–5 energies); and forensic analysis of ladder losses, which found the replay misalignment, a module-name collision that crashed a submission, belief-state carry-over between games of a match, and deck-search choices blind to evolution lines. The recurrent policy tied the memoryless one (47.5%, CI [32.9, 62.5], 40 games); the asymmetric (oracle) critic did not fit values better than the observable critic.

Appendix B: additional second-generation results

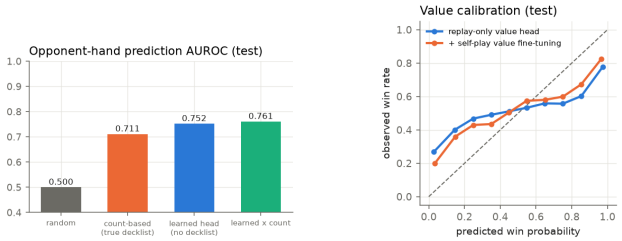


Figure 2. Left: opponent-hand prediction AUROC on held-out games. Right: value calibration before and after self-play fine-tuning.

Training and compute. Supervised training: AdamW (lr $4 \cdot 10^{-4}$, weight decay 0.05, batch 512, 12 epochs, loss weights policy 1, value 0.25, count 0.2, belief 0.05; moves of the losing player weighted 0.5), ≈45 min on one RTX 4070 Ti SUPER. Value fine-tuning and ExIt: 3 epochs, lr $1.5 \cdot 10^{-4}$, on 8k policy self-play games (value) and on 2,400 MCTS-16 / 1,600 MCTS-64 self-play games (ExIt rounds). Evaluation: 20 games in parallel on a 24-thread CPU, one CPU thread per agent, with Kaggle's 600 s bank simulated per game (it resets each game of a best-of-3 match on the ladder); the deployed agent averages ≈2.4 s per move.

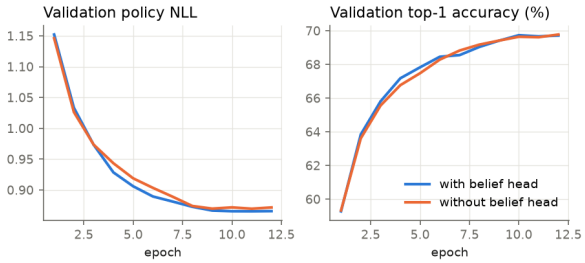


Figure 3. Validation curves with and without the auxiliary belief head (no effect on the policy).

Table 4. Further ablations (Crustle mirror; score of A vs B, 95% CI).

learned vs count-based hands, 256 sims (208)	.52 [.45, .58]
deployed: 1 vs 4 worlds (400)	.57 [.52, .61]
$c_{\text{puct}} = 0.7$ vs 1.25 (240)	.51 [.45, .58]
$c_{\text{puct}} = 2.5$ vs 1.25 (240)	.43 [.37, .49]
3 s cap + 120-deck library vs 2 s, 8 decks (160)	.54 [.46, .61]

Table 5. Deck selection: the Gen-2 policy piloting each deck against the gauntlet (240 games each).

Mega Kangaskhan ex / Crustle	.73 [.67, .78]
Teal Mask Ogerpon ex	.65 [.59, .71]
Hydrapple ex	.64 [.58, .70]
Marnie's Grimmsnarl ex	.59 [.53, .65]
Dragapult ex	.49 [.43, .56]
Lillie's Clefairy ex	.28 [.23, .34]